Unsupervised Pipeline High-level Description

Phase 0: Data Integrity & Validation

Key Question - how does the data look?

1. Combine CXR meta and chexpert files, filter on pneumonia (0,1) and jpeg image position (Chest PA and Lat)
2. Check for missing/null values, image paths, etc.

Phase 1 - Preprocessing & KMeans Clustering Run (centroid based, spherical shapes)

Key Question - does K-means do a good job of clustering pneumonia vs. not?

1. Using HuggingFace transform and pre-trained ResNet50: https://huggingface.co/Lab-Rasool/RadImageNet
2. Extract embeddings from ResNet50 for each image to use for unsupervised task instead of using raw pixels
3. Standardize embeddings using StandardScaler and use PCA for dimensionality reduction fixed to 50 components to create a stable baseline.
4. Use PCA space for downstream KMeans, DBSCAN and visualizations
5. Conduct a KMeans sweep across a range of predefined clusters from 2 to 10 and evaluate model for silhouette score, ARI and NMI. Goal to find high silhouette and high ARI/NMI.
6. Visualize using tSNE computed from PCA features and compare clusters with pneumonia classification.

Phase 2 - Preprocessing & DBSCAN Clustering Run (density based, complex shapes, noise/outlier discovery)

Key Question - are there complex clusters that DBSCAN can identify for pneumonia vs. not?

1. PCA features same as above for KMeans
2. Optimize parameters: eps (using k-distance plot) and min-samples over a range (5, 10, 20)
3. Evalute how much was assigned noise/outliers and if they are not pneumonia.

Phase 3 - Pneumonia Phenotypes (if time permits)

1. Restrict Pneumonia flag = 1 and same steps as above: PCA, KMeans, DBSCAN, Evaluation, Visualization

00 - Imports, Dataset, Save Along the Way

In [13]:
```
import os
import numpy as np
import pandas as pd
from pathlib import Path
import torch
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader
from torchvision import transforms, models
from PIL import Image
from tqdm.auto import tqdm
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.decomposition import PCA
import joblib
import matplotlib.pyplot as plt
```

In [2]:
```
ARTIFACTS_DIR = Path("artifacts")
RESULTS_DIR = Path("results")
FIG_DIR = Path("figures")

for d in [ARTIFACTS_DIR, RESULTS_DIR, FIG_DIR]:
    d.mkdir(parents=True, exist_ok=True)

IMAGE_COL = "image_path"
LABEL_COL = "pneumonia"
```

In [3]:
```
chex_img_df = pd.read_csv('chex_img_merged.csv')

df = chex_img_df.copy()

df = df.dropna(subset=["image_path", "pneumonia"]).copy()
df["pneumonia"] = df["pneumonia"].astype(int)
df = df.drop_duplicates(subset=["image_path"]).reset_index(drop=True)

print("N:", len(df))
print(df["pneumonia"].value_counts())
```

```
N: 69580
0    43358
1    26222
Name: pneumonia, dtype: int64
```

01 - Extract Embeddings for PCA

In [4]:
```
preprocess = transforms.Compose([
    transforms.Resize(256),
    transforms.CenterCrop(224),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406],
                         std=[0.229, 0.224, 0.225])
])
```

In [5]:
```
class CXRDataset(Dataset):
    def __init__(self, df, image_col="image_path", label_col="pneumonia", transform=N
        self.df = df.reset_index(drop=True)
        self.image_col = image_col
        self.label_col = label_col
        self.transform = transform or preprocess

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        row = self.df.iloc[idx]
        path = row[self.image_col]
        y = int(row[self.label_col])

        img = Image.open(path).convert("RGB")   # important
        x = self.transform(img)
        return x, y, idx
```

In [6]:
```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
device
```

Out[6]:
```
device(type='cuda')
```

In [7]:
```python
from torchvision import models
import torch.nn as nn

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

model = models.resnet50(weights=models.ResNet50_Weights.DEFAULT)

# Remove classification head → output 2048-d embeddings
model.fc = nn.Identity()

model.eval().to(device)
```

Out[7]:
```
ResNet(
  (conv1): Conv2d(3, 64, kernel_size=(7, 7), stride=(2, 2), padding=(3, 3), bias=Fals
e)
  (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=Tr
ue)
  (relu): ReLU(inplace=True)
  (maxpool): MaxPool2d(kernel_size=3, stride=2, padding=1, dilation=1, ceil_mode=Fals
e)
  (layer1): Sequential(
    (0): Bottleneck(
      (conv1): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stat
s=True)
      (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias
=False)
      (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stat
s=True)
      (conv3): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (relu): ReLU(inplace=True)
      (downsample): Sequential(
        (0): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (1): Bottleneck(
      (conv1): Conv2d(256, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stat
s=True)
      (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias
=False)
      (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stat
s=True)
      (conv3): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (relu): ReLU(inplace=True)
    )
    (2): Bottleneck(
      (conv1): Conv2d(256, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stat
s=True)
      (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias
=False)
      (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stat
s=True)
      (conv3): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (relu): ReLU(inplace=True)
    )
  )
  (layer2): Sequential(
    (0): Bottleneck(
      (conv1): Conv2d(256, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_sta
```

```
ts=True)
      (relu): ReLU(inplace=True)
      (downsample): Sequential(
        (0): Conv2d(256, 512, kernel_size=(1, 1), stride=(2, 2), bias=False)
        (1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (1): Bottleneck(
      (conv1): Conv2d(512, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (relu): ReLU(inplace=True)
    )
    (2): Bottleneck(
      (conv1): Conv2d(512, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (relu): ReLU(inplace=True)
    )
    (3): Bottleneck(
      (conv1): Conv2d(512, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (relu): ReLU(inplace=True)
    )
  )
  (layer3): Sequential(
    (0): Bottleneck(
      (conv1): Conv2d(512, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (relu): ReLU(inplace=True)
      (downsample): Sequential(
        (0): Conv2d(512, 1024, kernel_size=(1, 1), stride=(2, 2), bias=False)
        (1): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      )
```

```
    )
    (1): Bottleneck(
      (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (relu): ReLU(inplace=True)
    )
    (2): Bottleneck(
      (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (relu): ReLU(inplace=True)
    )
    (3): Bottleneck(
      (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (relu): ReLU(inplace=True)
    )
    (4): Bottleneck(
      (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (relu): ReLU(inplace=True)
    )
    (5): Bottleneck(
      (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (relu): ReLU(inplace=True)
    )
```

```
  )
  (layer4): Sequential(
    (0): Bottleneck(
      (conv1): Conv2d(1024, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(512, 2048, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (relu): ReLU(inplace=True)
      (downsample): Sequential(
        (0): Conv2d(1024, 2048, kernel_size=(1, 1), stride=(2, 2), bias=False)
        (1): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      )
    )
    (1): Bottleneck(
      (conv1): Conv2d(2048, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(512, 2048, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (relu): ReLU(inplace=True)
    )
    (2): Bottleneck(
      (conv1): Conv2d(2048, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bi
as=False)
      (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (conv3): Conv2d(512, 2048, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn3): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      (relu): ReLU(inplace=True)
    )
  )
  (avgpool): AdaptiveAvgPool2d(output_size=(1, 1))
  (fc): Identity()
)
```

In [12]:

```
#extract embeddings and save them
emb_path = ARTIFACTS_DIR / "embeddings.npy"
lab_path = ARTIFACTS_DIR / "labels.npy"
meta_path = ARTIFACTS_DIR / "meta.csv"

if emb_path.exists() and lab_path.exists() and meta_path.exists():
    X = np.load(emb_path)
    y = np.load(lab_path)
    meta = pd.read_csv(meta_path)
    print("Loaded cached embeddings:", X.shape)

else:
    ds = CXRDataset(df, image_col=IMAGE_COL, label_col=LABEL_COL, transform=preproces

    dl = DataLoader(
```

```
        ds,
        batch_size=64,
        shuffle=False,
        num_workers=4,
        pin_memory=(device.type == "cuda"),
    )

    N = len(ds)
    X = np.zeros((N, 2048), dtype=np.float32)
    y = np.zeros((N,), dtype=np.int64)

    model.eval()
    with torch.no_grad():
        for xb, yb, idx in tqdm(dl, total=len(dl)):
            xb = xb.to(device, non_blocking=True)
            feats = model(xb).detach().cpu().numpy().astype(np.float32)
            idx_np = idx.numpy()
            X[idx_np] = feats
            y[idx_np] = yb.numpy()

    meta = df.reset_index(drop=True).copy()
    meta.to_csv(meta_path, index=False)

    np.save(emb_path, X)
    np.save(lab_path, y)

    print("Saved:", emb_path, lab_path, meta_path)
```

```
  0%|          | 0/1088 [00:00<?, ?it/s]
Saved: artifacts/embeddings.npy artifacts/labels.npy artifacts/meta.csv
```

02 - PCA Analysis

In [15]:

```
#review the embedding artifacts loaded from previous
X = np.load(ARTIFACTS_DIR / "embeddings.npy")
y = np.load(ARTIFACTS_DIR / "labels.npy")
meta = pd.read_csv(ARTIFACTS_DIR / "meta.csv")

X.shape, y.shape, meta.shape
```

Out[15]:

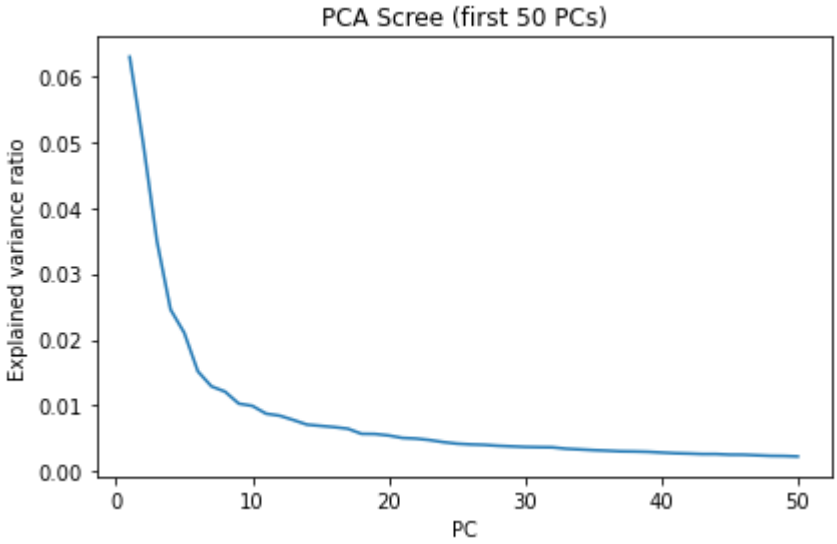
```
((69580, 2048), (69580,), (69580, 4))
```

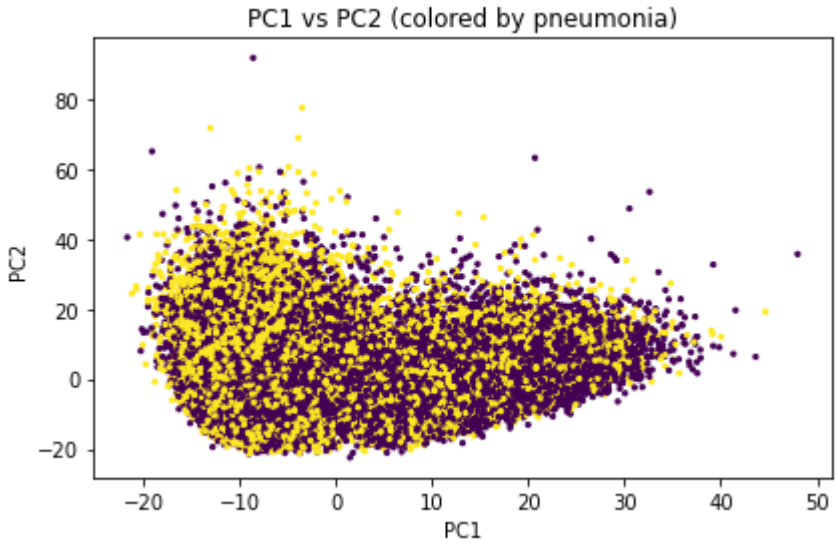
In [16]:

```
#standardize the embeddings
scaler = StandardScaler()
X_std = scaler.fit_transform(X)

pca = PCA(n_components=50, random_state=42)
X_pca = pca.fit_transform(X_std)

joblib.dump({"scaler": scaler, "pca": pca}, ARTIFACTS_DIR / "pca.joblib")
np.save(ARTIFACTS_DIR / "X_pca.npy", X_pca)

evr = pca.explained_variance_ratio_
pd.DataFrame({
    "component": np.arange(1, len(evr)+1),
    "explained_variance_ratio": evr,
    "cumulative": np.cumsum(evr)
}).to_csv(RESULTS_DIR / "pca_variance.csv", index=False)

print("Saved PCA artifacts.")
print("Cumulative variance @50:", np.cumsum(evr)[-1])
```

```
Saved PCA artifacts.
Cumulative variance @50: 0.42345948607822
```

In [17]:
```
#visualize the PCA components
# scree
plt.figure()
plt.plot(np.arange(1, len(evr)+1), evr)
plt.xlabel("PC")
plt.ylabel("Explained variance ratio")
plt.title("PCA Scree (first 50 PCs)")
plt.tight_layout()
plt.savefig(FIG_DIR / "pca_scree.png", dpi=200)

# PC1 vs PC2
plt.figure()
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y, s=5)
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("PC1 vs PC2 (colored by pneumonia)")
plt.tight_layout()
plt.savefig(FIG_DIR / "pca_pc1_pc2.png", dpi=200)
plt.show()
```





In [19]:
```
pc_means = []
for j in range(5):
    pc_means.append({
        "pc": j+1,
        "mean_pneumonia0": X_pca[y==0, j].mean(),
        "mean_pneumonia1": X_pca[y==1, j].mean(),
        "diff": X_pca[y==1, j].mean() - X_pca[y==0, j].mean()
    })
pc_means_df = pd.DataFrame(pc_means)
pc_means_df
```

Out[19]:

	pc	mean_pneumonia0	mean_pneumonia1	diff
0	1	0.867585	-1.434549	-2.302134
1	2	-0.711522	1.176499	1.888021
2	3	0.746677	-1.234628	-1.981305
3	4	0.069803	-0.115420	-0.185222
4	5	-0.723461	1.196241	1.919702

03 - KMeans Model Run

In [21]:
```
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, adjusted_rand_score, normalized_mutual_
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [24]:
```
X_pca = np.load(ARTIFACTS_DIR / "X_pca.npy")      # (N, D)
y = np.load(ARTIFACTS_DIR / "labels.npy")         # (N,)
```

In [25]:
```
rng = np.random.default_rng(42)
sample_size = min(5000, len(y))
sample_idx = rng.choice(len(y), size=sample_size, replace=False)

y_s = y[sample_idx].astype(np.int32)
```

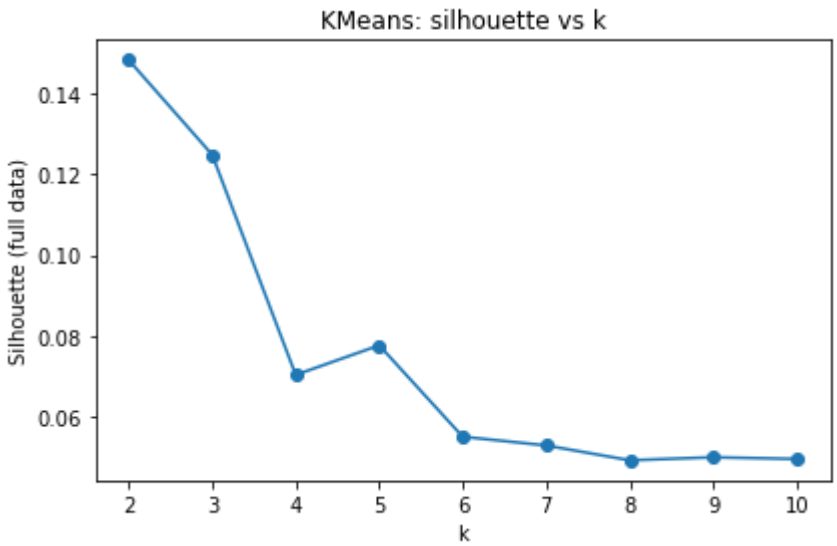
In [27]:
```
rows = []
k_range = range(2, 11)

for k in tqdm(k_range, desc="KMeans run"):
    km = KMeans(n_clusters=k, n_init=20, random_state=42)
    labels = km.fit_predict(X_pca).astype(np.int32)

    # Silhouette score on full dataset
    sil = silhouette_score(X_pca, labels)

    # ARI and NMI on sample to test a few
    labels_s = labels[sample_idx]
    ari = adjusted_rand_score(y_s, labels_s)
    nmi = normalized_mutual_info_score(y_s, labels_s)

    # Cluster diagnostics
    counts = np.bincount(labels)
    rows.append({
        "k": k,
        "silhouette": float(sil),
        "ari_sample": float(ari),
        "nmi_sample": float(nmi),
        "min_cluster": int(counts.min()),
        "max_cluster": int(counts.max()),
        "sample_size": int(sample_size),
    })

kmeans_run = pd.DataFrame(rows).sort_values("k")
kmeans_run.to_csv(RESULTS_DIR / "kmeans_run.csv", index=False)

display(kmeans_run)
```

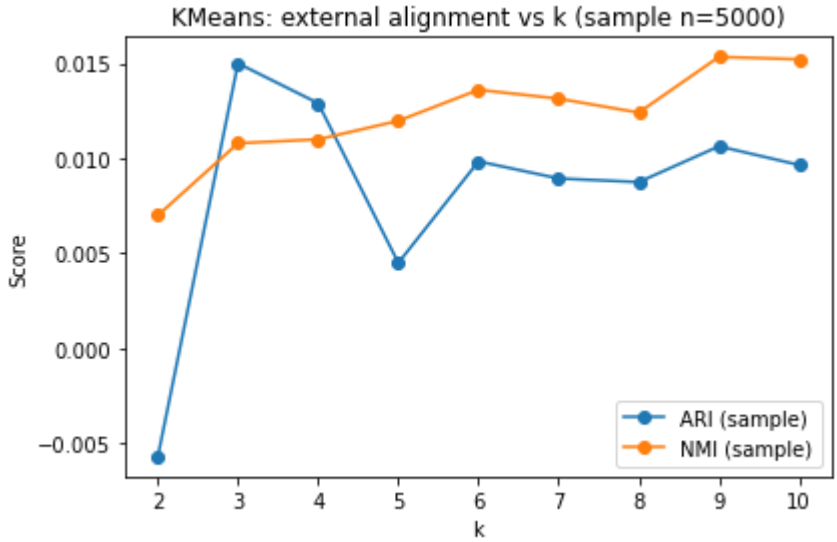
```
KMeans run:   0%|          | 0/9 [00:00<?, ?it/s]
```

	k	silhouette	ari_sample	nmi_sample	min_cluster	max_cluster	sample_size
0	2	0.148164	-0.005684	0.006998	24383	45197	5000
1	3	0.124773	0.014992	0.010791	13955	31777	5000
2	4	0.070416	0.012896	0.010994	10695	23396	5000
3	5	0.077715	0.004498	0.011970	7660	21671	5000
4	6	0.055150	0.009835	0.013602	6199	16673	5000
5	7	0.052997	0.008938	0.013136	5979	16322	5000
6	8	0.049328	0.008748	0.012403	3812	15101	5000
7	9	0.050078	0.010631	0.015334	3438	14110	5000
8	10	0.049656	0.009648	0.015206	3366	14028	5000

In [29]:

```
plt.figure()
plt.plot(kmeans_run["k"], kmeans_run["silhouette"], marker="o")
plt.xlabel("k")
plt.ylabel("Silhouette (full data)")
plt.title("KMeans: silhouette vs k")
plt.tight_layout()
plt.show()

plt.figure()
plt.plot(kmeans_run["k"], kmeans_run["ari_sample"], marker="o", label="ARI (sample)")
plt.plot(kmeans_run["k"], kmeans_run["nmi_sample"], marker="o", label="NMI (sample)")
plt.xlabel("k")
plt.ylabel("Score")
plt.title(f"KMeans: external alignment vs k (sample n={sample_size})")
plt.legend()
plt.tight_layout()
plt.show()
```





In [31]:
```
#pick the best k
# simple heuristic: maximize ARI, break ties with silhouette
best_row = kmeans_run.sort_values(["ari_sample", "silhouette"], ascending=False).iloc
best_k = int(best_row["k"])
print("Best k:", best_k, best_row.to_dict())

km = KMeans(n_clusters=best_k, n_init=20, random_state=42)
kmeans_labels = km.fit_predict(X_pca)

np.save(ARTIFACTS_DIR / f"kmeans_labels_k{best_k}.npy", kmeans_labels)
```

```
Best k: 3 {'k': 3.0, 'silhouette': 0.12477309256792068, 'ari_sample': 0.0149923293578
3689, 'nmi_sample': 0.01079079328294794, 'min_cluster': 13955.0, 'max_cluster': 3177
7.0, 'sample_size': 5000.0}
```

In [52]:
```
best_k
```

Out[52]: 3

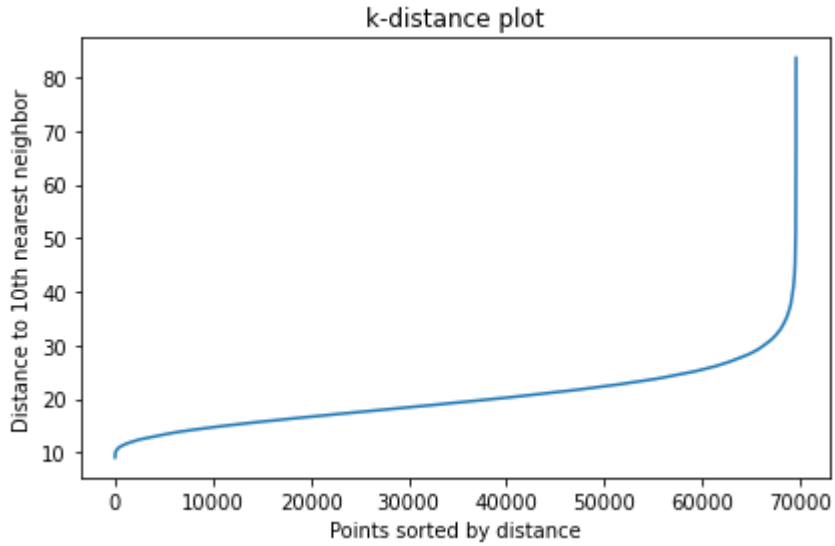
04 - DBSCAN Run

In [32]:
```
#imports and use saved data
from sklearn.cluster import DBSCAN
from sklearn.neighbors import NearestNeighbors
from sklearn.metrics import silhouette_score, adjusted_rand_score, normalized_mutual_
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

X_pca = np.load(ARTIFACTS_DIR / "X_pca.npy")
y = np.load(ARTIFACTS_DIR / "labels.npy")
```

In [ ]:
```
#https://github.com/amueller/mglearn/blob/main/mglearn/plot_dbscan.py
#https://medium.com/@kaleemullahyounas123/dbscan-another-clustering-algorithm-for-mac
```

In [36]:
```
#choosing eps and min_samples
def k_distance_plot(X, k=10):
    nn = NearestNeighbors(n_neighbors=k)
    nn.fit(X)
    distances, _ = nn.kneighbors(X)
    # distance to kth neighbor
    kth = np.sort(distances[:, -1])
    plt.figure()
    plt.plot(kth)
    plt.xlabel("Points sorted by distance")
    plt.ylabel(f"Distance to {k}th nearest neighbor")
```

```
    plt.title("k-distance plot")
    plt.tight_layout()
    plt.show()
    return kth

_ = k_distance_plot(X_pca, k=10)
```



In [46]:
```
#guess the elbow/knee based on k-distance plot above
elbow_pt = 30

#define the parameter grid
eps_grid = np.linspace(10, elbow_pt, 10)
min_samples_grid = [5, 10, 20]
```

In [47]:
```
#run a sample for the ARI/NMI
rng = np.random.default_rng(42)
sample_size = min(5000, len(y))
sample_idx = rng.choice(len(y), size=sample_size, replace=False)
y_s = y[sample_idx]
```

In [48]:
```
rows = []

for ms in tqdm(min_samples_grid, desc="DBSCAN min_samples"):
    for eps in tqdm(eps_grid, desc=f"eps grid (ms={ms})", leave=False):
        db = DBSCAN(eps=float(eps), min_samples=int(ms))
        labels = db.fit_predict(X_pca).astype(np.int32)

        non_noise = labels != -1
        noise_pct = float((~non_noise).mean())

        # number of clusters excluding noise
        n_clusters = int(len(set(labels[non_noise]))) if non_noise.any() else 0

        # ARI/NMI calculations
        labels_s = labels[sample_idx]
        ari = adjusted_rand_score(y_s, labels_s)
        nmi = normalized_mutual_info_score(y_s, labels_s)

        #Silhouette Calculations
        sil = np.nan
        if n_clusters >= 2 and non_noise.sum() > 50:
            try:
                sil = float(silhouette_score(X_pca[non_noise], labels[non_noise]))
            except Exception:
                sil = np.nan

        rows.append({
            "eps": float(eps),
```

```
            "min_samples": int(ms),
            "n_clusters": n_clusters,
            "noise_pct": noise_pct,
            "ari_sample": float(ari),
            "nmi_sample": float(nmi),
            "silhouette_non_noise": sil,
            "sample_size": int(sample_size),
        })

dbscan_sweep = pd.DataFrame(rows)
dbscan_sweep.to_csv(RESULTS_DIR / "dbscan_sweep.csv", index=False)

display(dbscan_sweep.sort_values(["ari_sample", "n_clusters", "silhouette_non_noise"]
```

```
DBSCAN min_samples:   0%|          | 0/3 [00:00<?, ?it/s]
eps grid (ms=5):   0%|          | 0/10 [00:00<?, ?it/s]
eps grid (ms=10):   0%|          | 0/10 [00:00<?, ?it/s]
eps grid (ms=20):   0%|          | 0/10 [00:00<?, ?it/s]
```

	eps	min_samples	n_clusters	noise_pct	ari_sample	nmi_sample	silhouette_non_noi
6	23.333333	5	1	0.126832	0.009165	0.001273	N
17	25.555556	10	1	0.074619	0.009150	0.001640	N
7	25.555556	5	1	0.070652	0.008817	0.001607	N
27	25.555556	20	1	0.078916	0.008689	0.001429	N
16	23.333333	10	1	0.133659	0.008292	0.001038	N
26	23.333333	20	1	0.142110	0.007881	0.000931	N
5	21.111111	5	3	0.226847	0.004159	0.000315	0.1462
28	27.777778	20	1	0.043533	0.003746	0.000469	N
25	21.111111	20	2	0.252271	0.003302	0.000397	0.2082
15	21.111111	10	1	0.239451	0.003249	0.000211	N
18	27.777778	10	1	0.041190	0.002093	0.000157	N
8	27.777778	5	2	0.038919	0.001735	0.000115	0.2813
21	12.222222	20	2	0.943806	0.001631	0.001520	0.2353
9	30.000000	5	1	0.021026	0.001448	0.000135	N
19	30.000000	10	1	0.022190	0.001389	0.000122	N
1	12.222222	5	10	0.920509	0.000940	0.002086	-0.0412
24	18.888889	20	1	0.416715	0.000643	0.000106	N
14	18.888889	10	2	0.397140	0.000612	0.000218	0.1902
29	30.000000	20	1	0.023642	0.000549	0.000019	N
4	18.888889	5	3	0.379218	0.000414	0.000177	0.1276

In [50]:

```
#choose the best DBSCAN parameters from the grid earlier
best = dbscan_sweep.sort_values(["ari_sample", "n_clusters", "silhouette_non_noise"],
best_eps = float(best["eps"])
best_ms = int(best["min_samples"])
print("Best DBSCAN:", best.to_dict())

db = DBSCAN(eps=best_eps, min_samples=best_ms)
dbscan_labels = db.fit_predict(X_pca)

np.save(ARTIFACTS_DIR / "dbscan_labels_best.npy", dbscan_labels)
```

```
Best DBSCAN: {'eps': 23.333333333333336, 'min_samples': 5.0, 'n_clusters': 1.0, 'nois
e_pct': 0.1268324231100891, 'ari_sample': 0.009165001727586046, 'nmi_sample': 0.00127
26491163974139, 'silhouette_non_noise': nan, 'sample_size': 5000.0}
```

05 - tSNE for visualization

In [51]:
```
from sklearn.manifold import TSNE
```

In [53]:
```
# KMeans: use whichever best_k you saved earlier
kmeans_labels = np.load(ARTIFACTS_DIR / f"kmeans_labels_k{best_k}.npy").astype(int)

# DBSCAN load
dbscan_labels = np.load(ARTIFACTS_DIR / "dbscan_labels_best.npy").astype(int)

X_pca.shape, y.shape, kmeans_labels.shape, dbscan_labels.shape
```

Out[53]:
```
((69580, 50), (69580,), (69580,), (69580,))
```

In [54]:
```
#run tSNE on a sample
rng = np.random.default_rng(42)
N = X_pca.shape[0]
tsne_n = min(5000, N)  # 5k is usually plenty for visualization
idx = rng.choice(N, size=tsne_n, replace=False)

X_tsne_in = X_pca[idx]
y_s = y[idx]
kmeans_s = kmeans_labels[idx]
dbscan_s = dbscan_labels[idx]

tsne_out_path = ARTIFACTS_DIR / f"tsne_2d_n{tsne_n}.npy"
idx_path = ARTIFACTS_DIR / f"tsne_idx_n{tsne_n}.npy"
```

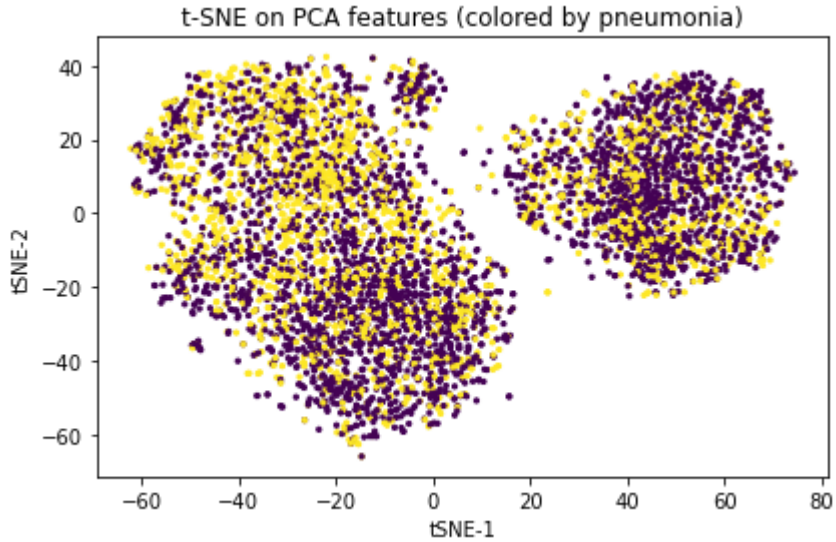
In [60]:
```
dbscan_s
```

Out[60]:
```
array([ 0,  0, -1, ...,  0, -1,  0])
```

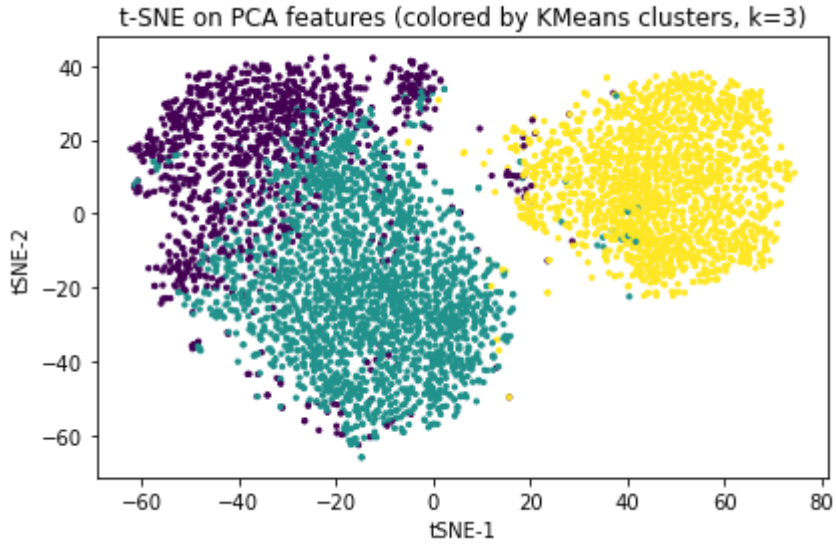
In [63]:
```
#load tSNE
if tsne_out_path.exists() and idx_path.exists():
    Z = np.load(tsne_out_path)
    idx_loaded = np.load(idx_path)
    # sanity check that we're using same subset
    assert len(idx_loaded) == len(idx)  # if this fails, delete cached files and reru
    print("Loaded cached t-SNE:", Z.shape)
else:
    tsne = TSNE(
        n_components=2,
        perplexity=30,
        learning_rate=200,
        init="pca",
        random_state=42,
        n_iter=1000,
    )
    Z = tsne.fit_transform(X_tsne_in)

    np.save(tsne_out_path, Z)
    np.save(idx_path, idx)
    print("Saved t-SNE:", tsne_out_path)
```

```
Saved t-SNE: artifacts/tsne_2d_n5000.npy
```

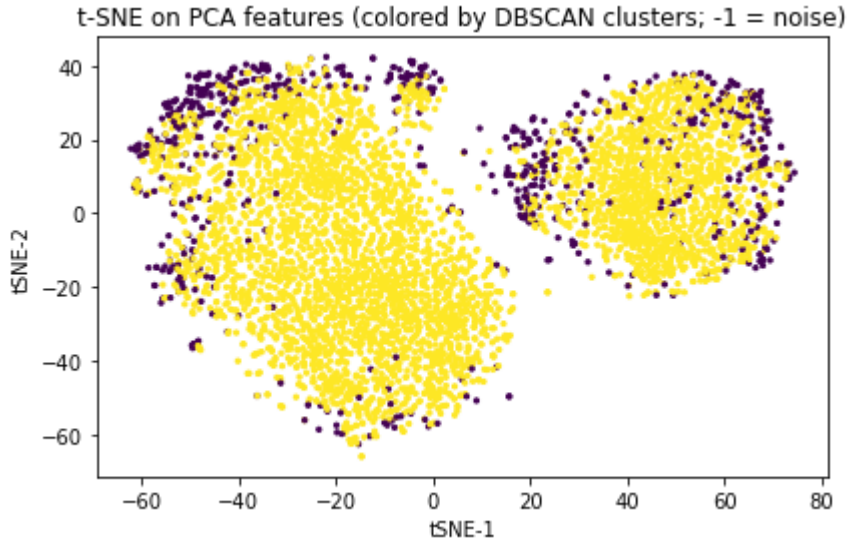
In [65]:
```
#tSNE w pneumonia labels shown
plt.figure()
plt.scatter(Z[:, 0], Z[:, 1], c=y_s, s=5)
plt.title("t-SNE on PCA features (colored by pneumonia)")
plt.xlabel("tSNE-1")
plt.ylabel("tSNE-2")
plt.tight_layout()
plt.show()
```



In [66]:
```
#tSNE w KMeans clusters shown
plt.figure()
plt.scatter(Z[:, 0], Z[:, 1], c=kmeans_s, s=5)
plt.title(f"t-SNE on PCA features (colored by KMeans clusters, k={best_k})")
plt.xlabel("tSNE-1")
plt.ylabel("tSNE-2")
plt.tight_layout()
plt.show()
```



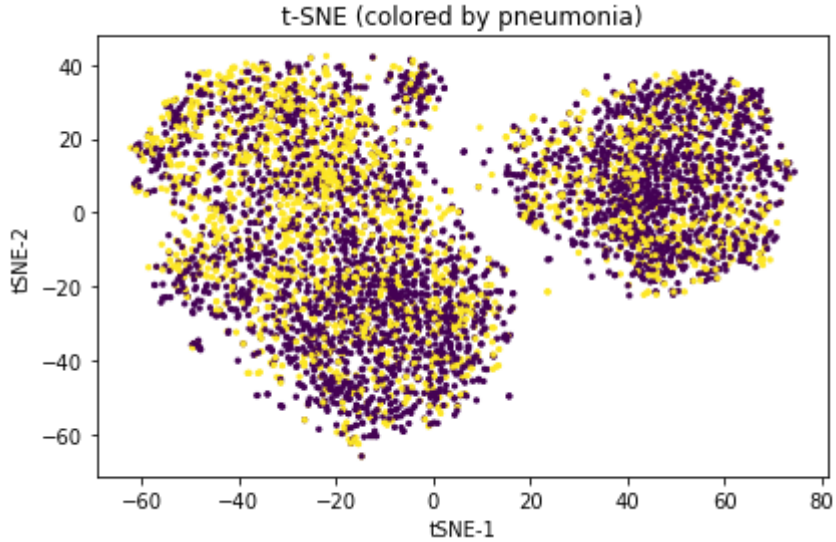
In [67]:
```
# remap labels so noise (-1) becomes 0 and clusters shift by +1 (purely for plotting)
dbscan_plot = dbscan_s.copy()
dbscan_plot = np.where(dbscan_plot == -1, -1, dbscan_plot)

plt.figure()
plt.scatter(Z[:, 0], Z[:, 1], c=dbscan_plot, s=5)
plt.title("t-SNE on PCA features (colored by DBSCAN clusters; -1 = noise)")
plt.xlabel("tSNE-1")
plt.ylabel("tSNE-2")
plt.tight_layout()
plt.show()

print("DBSCAN noise % in t-SNE subset:", (dbscan_s == -1).mean())
```

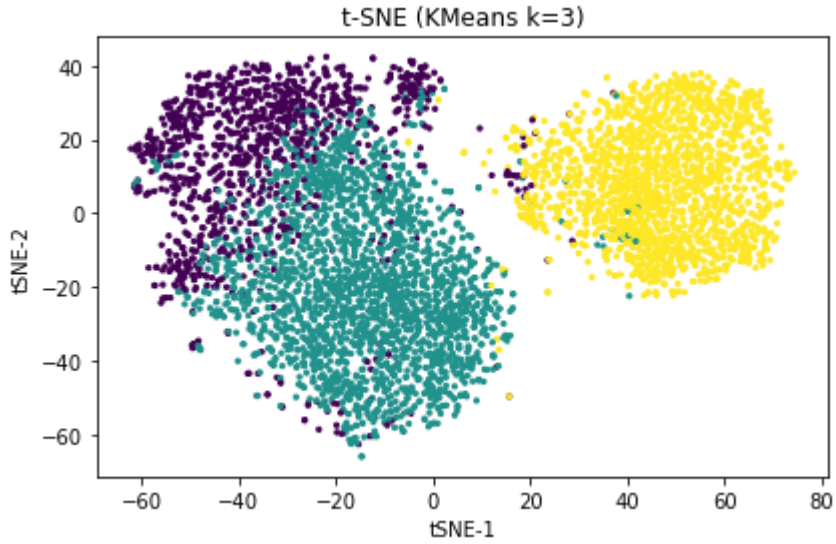


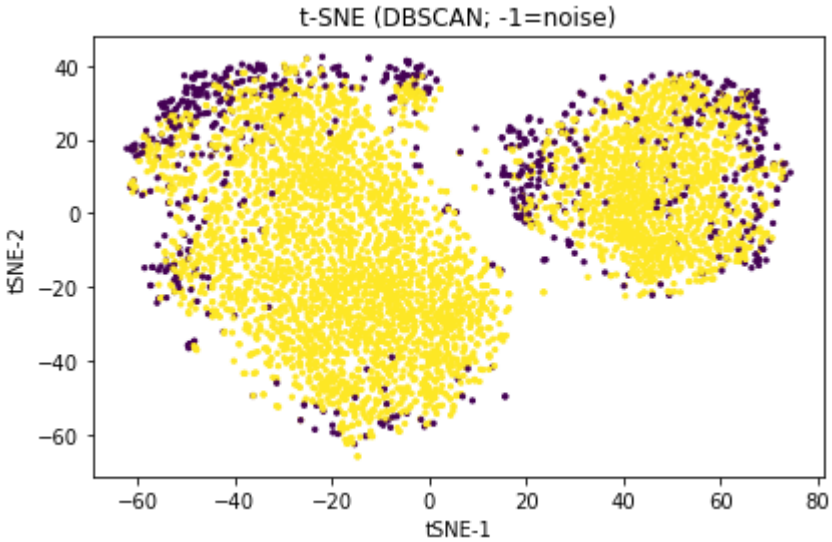

```
DBSCAN noise % in t-SNE subset: 0.1344
```

In [68]:

```
def save_scatter(Z, c, title, filename):
    plt.figure()
    plt.scatter(Z[:, 0], Z[:, 1], c=c, s=5)
    plt.title(title)
    plt.xlabel("tSNE-1")
    plt.ylabel("tSNE-2")
    plt.tight_layout()
    plt.savefig(FIG_DIR / filename, dpi=200)
    plt.show()

save_scatter(Z, y_s, "t-SNE (colored by pneumonia)", f"tsne_pneumonia_n{tsne_n}.png")
save_scatter(Z, kmeans_s, f"t-SNE (KMeans k={best_k})", f"tsne_kmeans_k{best_k}_n{tsn
save_scatter(Z, dbscan_plot, "t-SNE (DBSCAN; -1=noise)", f"tsne_dbscan_n{tsne_n}.png"
```







In [70]:

```
# Rough split by tSNE-1 coordinate
left = Z[:, 0] < np.median(Z[:, 0])
right = Z[:, 0] >= np.median(Z[:, 0])

print("Left pneumonia rate:", y_s[left].mean())
print("Right pneumonia rate:", y_s[right].mean())
```

```
Left pneumonia rate: 0.438
Right pneumonia rate: 0.328
```